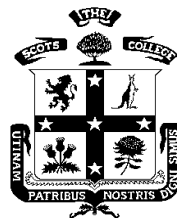


# The Scots College



## 2009 Trial HSC Examination

### Year 12 Mathematics

### Extension 1

#### General Instructions

- Reading time: 5 mins
- Working time : 2 Hours
- Write using a blue or black pen
- Board approved calculators may be used
- A standard table of integrals is provided
- All necessary working should be shown in every question
- Start each question in a new booklet

Total Marks HSC: 84

Weighting : 40%

Attempt Questions 1—7

All questions are of equal value

**Question 1 ( 12 marks) Start in a SEPARATE booklet****Marks**

(a) Find  $\int \frac{1}{16x^2 + 1} dx$  (2)

(b) Using the substitution  $u = x^3 + 3x$ , or otherwise, find (3)

$$\int (x^2 + 1)(x^3 + 3x)^{-3} dx$$

(c) Solve  $\frac{2}{x-3} < 4$ . (3)

(d) Evaluate (2)

$$\lim_{x \rightarrow 0} \frac{4 \sin 2x}{3x}$$

(e) Let  $A$  be the point  $(-2, -2)$  and let  $B$  be the point  $(6, 10)$ . Find the coordinates of the point  $P$  which divides the interval  $AB$  externally in the ratio 1 : 2. (2)

**Question 2 ( 12 marks) Start in a SEPARATE booklet****Marks**

- (a) Find (2)

$$\frac{d}{dx} (x \sin^{-1} 3x)$$

- (b) Let  $f(x) = 2x^2 + 3x$ . Use the definition (2)

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

to find the derivative of  $f(x)$  at the point  $x = 5$ .

- (c) Find (3)

$$\int_0^{\pi/4} \cos^2 5x \, dx$$

- (d) Solve the equation (2)

$$2 \sin^2 \theta = 9 - 15 \cos \theta, \quad 0^\circ \leq \theta \leq 360^\circ$$

- (e) Use the principle of mathematical induction to show that  $x^n - y^n$  is divisible by  $x - y$  when  $n$  is a positive integer. (3)

**Question 3 ( 12 marks) Start in a SEPARATE booklet****Marks**

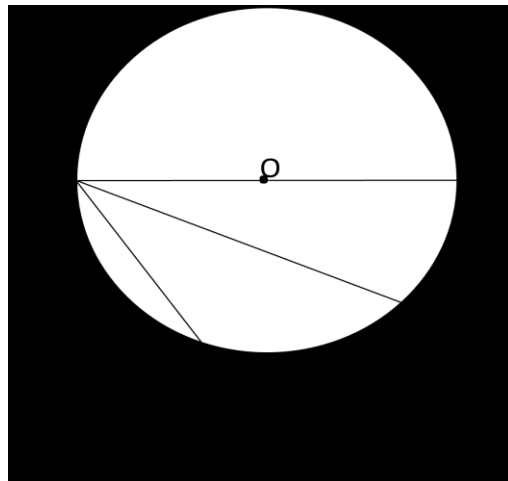
- (a) (i) Show that the equation  $\log_e x \times \sec x = 1$  has a root between 1 and 2. (1)

- (ii) Using the above and one application of Newton's method find a better approximation of the root of the equation in part (i), leaving your answer correct to three decimal places. (3)

- (b) (i) Differentiate  $y = 5e^{5x} \sin x - e^{5x} \cos x$  with respect to  $x$ . (2)

- (ii) Hence, or otherwise, find  $\int e^{5x} \sin x \, dx$ . (1)

- (c) The figure below is not to scale.



NP is a tangent to the circle with centre O and LM is the diameter. If NM bisects the angle LNP, prove that MNP is a right angled triangle. (3)

- (d) Find the exact value of  $\sin\left(2 \cos^{-1}\left(\frac{3}{5}\right)\right)$  (2)

**Question 4 ( 12 marks) Start in a SEPARATE booklet****Marks**

- (a) A curve has parametric equations (2)

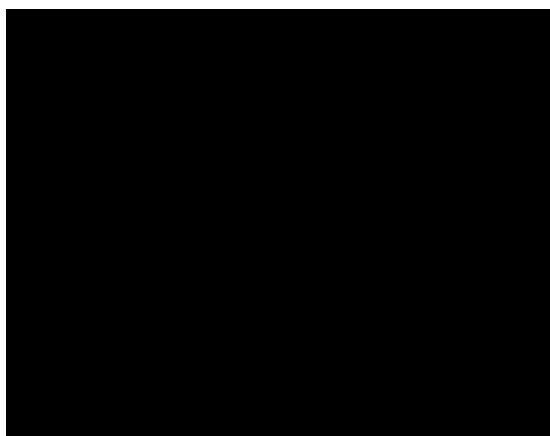
$$x = t - \frac{1}{t} \quad , \quad y = t^2 + 1 + \frac{1}{t^2}$$

Find the Cartesian equation for this curve.

- (b) (i) Express  $3 \sin x - 4 \cos x$  in the form  $R \sin(x - \alpha)$ . (3)  
Where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ .

- (ii) Hence or otherwise find the general solution of  $4 \cos x - 3 \sin x = 2.5$  (2)

- (c)



The conical vessel in the diagram is being filled with water at the rate of  $10 \text{ cm}^3/\text{min}$ . The semi-vertical angle of the vessel is  $45^\circ$ . The radius of the water surface is  $r \text{ cm}$  at any time  $t$  minutes, the perpendicular height is  $h \text{ cm}$  and the slant edge is  $l \text{ cm}$ .

- (i) Find the volume  $V$  of water in terms of the slant edge  $l$ . (2)

- (ii) Find the rate at which the slant edge  $l$  is increasing when the radius of the water surface is  $7 \text{ cm}$ . (3)

**Question 5 ( 12 marks) Start in a SEPARATE booklet****Marks**

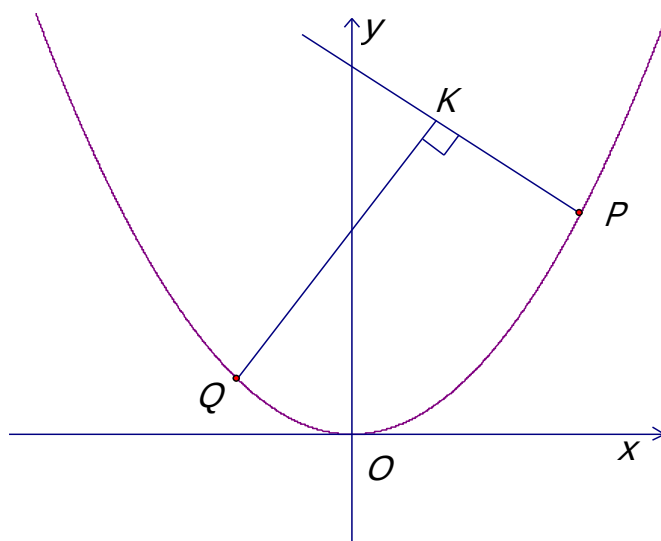
- (a) For the function  $y = 2 \cos^{-1}(4x - 1)$
- (i) State the largest possible domain and range. (2)
- (ii) Sketch a neat graph of the function. (2)
- (b) Aaron and David roll a die alternately in a game. The first person to roll a 6 wins the game. Aaron rolls first.
- (i) What is the probability of Aaron winning in his second roll? (1)
- (ii) What is the probability of Aaron winning the game? (2)
- (c) Solve the equation  $2x^3 - 7x^2 - x - 1 = 0$  , given that the roots are in geometric progression. (3)
- (d) If  $\cos ( 2 \sin x ) = \sin( 2 \cos x )$  , then show that (2)

$$\sin x \pm \cos x = n\pi \pm \frac{\pi}{4}$$

Question 6 ( 12 marks) Start in a SEPARATE booklet

Marks

(a)



$P ( 2at , at^2 )$  is any point on the parabola  $x^2 = 4ay$ .

(i) show that the equation of the Normal at  $P$  is (2)

$$y = at^2 + 2a - \frac{x}{t}$$

(ii)  $Q$  is another point on the parabola such that the Normals at  $P$  and  $Q$  are at right angles to each other and intersect at  $K$ . Find the coordinates of the point  $Q$  in terms of  $t$ . (2)

(iii) Show that the coordinates of  $K$  are (3)

$$x = at - \frac{a}{t} \quad \text{and} \quad y = at^2 + a + \frac{1}{t^2}.$$

**Question 6 continued.....**

- (b) The time is 10:20 a.m. and Arun has just finished making two cups of coffee which have a temperature of  $95^{\circ}\text{C}$ . The coffees are losing heat according to Newton's law of cooling  $\frac{dT}{dt} = -k(T - 20)$  where  $T$  is the temperature of the coffee in degrees Celsius after  $t$  minutes,  $k$  is a positive constant and the temperature of the room is  $20^{\circ}\text{C}$ .

(i) Show that  $T = 20 + 75e^{-kt}$  is a solution of the equation . (1)

(ii) At 10:30 a.m. the coffees have cooled to  $75^{\circ}\text{C}$ . Find the value of  $k$  to two decimal places. (2)

(iii) The second cup of coffee is for Lawrie who will drink it as long as its temperature is not less than  $60^{\circ}\text{C}$ . What is the latest time that he could arrive to drink his coffee? (2)

**Question 7 ( 12 marks) Start in a SEPARATE booklet****Marks**

- (a) In a scientific cloud seeding experiment a projectile is fired from  $O$  with a velocity  $V$  at angle of  $\theta$  to level ground. The cloud is at a height  $h$  above the ground. The projectile returns to the ground at a horizontal distance of  $Z$  metres.

The equations of motion of the projectile are

$$x = Vt\cos\theta, \quad y = -\frac{1}{2}gt^2 + Vt\sin\theta$$

(You are NOT Required to prove the above equations)

- (i) Show that when the projectile returns to the ground level (2)

$$Z = \frac{V^2 \sin 2\theta}{g} \text{ metres.}$$

- (ii) If the projectile MUST pass through the cloud twice then show that (2)

$$h < \frac{V^2 \sin^2 \theta}{2g}$$

- (iii) Consider  $g = 10 \frac{m}{s^2}$  ,  $\theta = 60^\circ$  ,  $V = 320 \frac{m}{s}$  and  $h = 3000 \text{ m}$ . (2)  
Find the times at which the projectile passes through the clouds. Leave your answer correct to one decimal.

- (iv) Find the horizontal distance covered between the two times found in part (iii). (2)

**Question 7 continued.....**

(b) A particle is moving so that  $\ddot{x} = 32x^3 - 48x^2 + 16x$

Initially the velocity is  $24 \text{ m/s}$  and  $x = 3$  metres.

(i) Show that  $v^2 = 16x^2 (x - 1)^2$  (2)

(ii) Hence or otherwise show that (2)

$$\int \frac{1}{x(x-1)} dx = 4t, \text{ where } t \text{ is time elapsed.}$$

## Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right) + C$$

**NOTE :**  $\ln x \equiv \log_e x, \quad x > 0$

Q1

(a)

$$\begin{aligned} \int \frac{1}{16x^2+1} dx &= \frac{1}{16} \int \frac{1}{x^2 + (\frac{1}{4})^2} dx \\ &= \frac{1}{16} \times \frac{1}{\frac{1}{4}} \tan^{-1}(4x) + C \\ &= \frac{1}{4} \tan^{-1}(4x) + C \end{aligned}$$

(b)  $\int (x^2+1)(x^3+3x)^{-3} dx$

$$\begin{aligned} u &= x^3+3x \\ \frac{du}{dx} &= 3x^2+3 \\ dx &= \frac{du}{3(x^2+1)} \end{aligned}$$

$$\therefore \int (x^2+1) u^{-3} \frac{du}{3(x^2+1)}$$

$$\Rightarrow \frac{1}{3} \frac{u^{-2}}{(-2)} + C$$

$$\Rightarrow -\frac{1}{6(x^3+3x)} + C$$

(c)

$$\frac{2}{x-3} < 4$$

$$2(x-3) < 4(x-3)^2$$

$$4(x-3)^2 - 2(x-3) > 0$$

$$2(x-3)[2(x-3)-1] > 0$$

$$2(x-3)(2x-7) > 0$$

$$x < 3 \quad \text{or} \quad x > \frac{7}{2}$$

(d)  $\lim_{x \rightarrow 0} 4 \frac{\sin 2x}{3x}$

$$= \frac{4}{3} \times 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$

$$= \frac{8}{3}$$

(e)

$$\begin{matrix} m:n \\ 1:-2 \end{matrix}$$

$$(-2, -2) \quad (6, 10)$$

$$\frac{mx_2 + nx_1}{m+n} \quad \frac{my_2 + ny_1}{m+n}$$

$$\frac{1(6) - 2(-2)}{1-2} \quad \frac{1(10) - 2(-2)}{1-2}$$

$$= -10 \quad = -14$$

$$(-10, -14)$$

Q.2

(a)

$$\begin{aligned} \frac{d}{dx} (x \sin^{-1} 3x) &= \sin^{-1} 3x (1) + x \frac{1}{\sqrt{1-9x^2}} (3) \\ &= \sin^{-1} 3x + \frac{3x}{\sqrt{1-9x^2}} \end{aligned}$$

(b)

$$\begin{aligned} f(a) &= 2a^2 + 3a \\ f(a+h) &= 2(a+h)^2 + 3(a+h) \\ &= 2a^2 + 4ah + 2h^2 + 3a + 3h \end{aligned}$$

$$f'(a) = \frac{2a^2 + 4ah + 2h^2 + 3a + 3h - 2a^2 - 3a}{h}$$

$$= 4a + 3$$

$$f'(5) = 4 \times 5 + 3 = 23$$

(c)

$$\int_0^{\pi/4} \cos^2 5x \, dx$$

$$\int_0^{\pi/4} \frac{1}{2} + \frac{1}{2} \cos 10x \, dx$$

$$\left[ \frac{x}{2} + \frac{1}{2} \sin 10x \times \frac{1}{10} \right]_0^{\pi/4}$$

$$\left[ \frac{x}{2} + 5 \sin 10x \right]_0^{\pi/4}$$

$$\frac{\pi}{8} + 5 \sin \frac{5\pi}{2} - 0 - 5 \times 0$$

$$\frac{\pi}{8} + 5 \sin \frac{5\pi}{2}$$

(d)

$$2 \sin^2 \theta = 9 - 15 \cos \theta$$

$$2(1 - \cos^2 \theta) = 9 - 15 \cos \theta$$

$$2 - 2 \cos^2 \theta = 9 - 15 \cos \theta$$

$$2 \cos^2 \theta - 15 \cos \theta + 7 = 0$$

$$(2 \cos \theta - 1)(\cos \theta - 7) = 0$$

$$\cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = 7$$

$$60^\circ, 300^\circ$$

Q.2

e)

 $x^n - y^n$  is divisible by  $x - y$ (a) for  $n=1$   $x - y$  is divisible by  $x - y$  $\therefore$  True for  $n=1$ (b) Assume that it is true for  $n=k$ i.e.  $x^k - y^k$  is divisible by  $x - y$  (Hypothesis) — (1)

$$\begin{aligned} \text{consider } x^{k+1} - y^{k+1} &= x^k \cdot x - y^k \cdot y \\ &= x^k \cdot x - x^k \cdot y + x^k \cdot y - y^k \cdot y \quad * \\ &= x^k(x - y) + y(x^k - y^k) \end{aligned}$$

The first term is divisible by  $(x - y)$ The 2nd term is divisible by  $(x - y)$  due to statement/assumption (1).i.e.  $x^{k+1} - y^{k+1}$  is divisible by  $x - y$ (c) Hence by the principle of mathematical induction it is true for all positive integral values of  $n$ .

Q.3.

(a) If  $f(x) = \cos x - \log_e x$ Assume  $x_0 = 1.5$ 

$$f'(x) = -\sin x - \frac{1}{x}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1.5 - \frac{(-0.3347)}{-1.6642}$$

$$= 1.29888$$

$$= 1.299$$

g.  $f(x) = \log_e x \sec x - 1 =$ 

$$f'(x) = \log_e x \sec x \tan x + \sec x \cdot \frac{1}{x}$$

$$x_0 = 1.5$$

$$x_1 = 1.5 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1.5 - \frac{4.73}{90.25}$$

$$= 1.4476$$

$$= 1.448$$

$$\begin{aligned} \text{(b)(i) } y &= 5e^{5x} \sin x - e^{5x} \cos x \\ &= e^{5x} (5 \sin x - \cos x) \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= (5 \sin x - \cos x) 5e^{5x} + e^{5x} (5 \cos x + \sin x) \\ &= e^{5x} [25 \sin x - 5 \cos x + 5 \cos x + \sin x] \\ &= e^{5x} [26 \sin x] \quad \checkmark \end{aligned}$$

(ii) From part (i)

$$\int e^{5x} 26 \sin x dx = 5e^{5x} \sin x - e^{5x} \cos x + C$$

$$\int e^{5x} \sin x dx = \frac{e^{5x}}{26} (5 \sin x - \cos x) + C$$

Q.3.

(c)

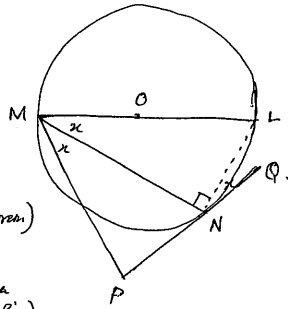
Construct LN

Let  $\angle LMN = \angle NMP = x$  [given] $\angle LNM = 90^\circ$  (Angle in a semicircle) $\angle LNQ = \angle NML = x$  (Alt. Ang. & Tangent Theorem) $\angle PNM = 180 - (90 + x)$  ( $\because$  PNQ is a straight line)  
 $\Rightarrow \angle PNM = 90 - x$ 

In triangle MNP

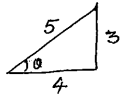
 $\angle MPN = 90^\circ$   $\because$  sum of angles in a triangle is  $180^\circ$ . $(x + 90 - x + \angle MPN = 180)$   
 $\angle MPN = 90^\circ$ 

Hence proved.



(a)  $\sin \left( 2 \cos^{-1} \left( \frac{3}{5} \right) \right)$

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \times \frac{4}{5} \times \frac{3}{5} \\ &= \frac{24}{25} \end{aligned}$$



Q.4.

(a)  $x = t - \frac{1}{t}, y = t^2 + 1 + \frac{1}{t^2}$

$$y = t^2 - 2t \times \frac{1}{t} + \frac{1}{t^2} + 1 + 2$$

$$y = \left( t - \frac{1}{t} \right)^2 + 3$$

$$y = x^2 + 3$$

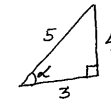
(b) (i)  $3 \sin x - 4 \cos x = R \sin(x - \alpha)$   
 $= R \sin x \cos \alpha - R \cos x \sin \alpha$

$$\Rightarrow R \cos \alpha = 3 \quad \text{--- (1)}$$

$$-R \sin \alpha = -4 \quad \text{--- (2)}$$

$$R^2 = 3^2 + 4^2$$

$$R = \pm 5$$



(2)  $\div$  (1)  $\Rightarrow \tan \alpha = \frac{4}{3}$

$$\alpha = \tan^{-1} \left( \frac{4}{3} \right) = 0.93^\circ \quad [53^\circ 8']$$

$$\therefore 3 \sin x - 4 \cos x = 5 \sin \left( x - \tan^{-1} \left( \frac{4}{3} \right) \right)$$

(ii)  $-5 \sin \left( x - \tan^{-1} \left( \frac{4}{3} \right) \right) = 2.5$

$$\sin \left( x - \tan^{-1} \left( \frac{4}{3} \right) \right) = -\frac{1}{2}$$

$$\sin \left( x - \tan^{-1} \left( \frac{4}{3} \right) \right) = -\sin \left( \frac{1}{2} \right)$$

$$x - \tan^{-1} \left( \frac{4}{3} \right) = n\pi + (-1)^n \left( -\sin^{-1} \left( \frac{1}{2} \right) \right)$$

$$x = n\pi + \tan^{-1} \left( \frac{4}{3} \right) + (-1)^n \left( -\sin^{-1} \left( \frac{1}{2} \right) \right)$$

(c)  $h = r$  since semi-vertical angle  $= 45^\circ$

$$l^2 = r^2 + r^2$$

$$l^2 = 2r^2$$

$$r = \frac{l}{\sqrt{2}}$$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^3$$

$$= \frac{1}{3} \pi \left( \frac{l}{\sqrt{2}} \right)^3$$

$$\frac{dV}{dl} = \frac{\pi l^2}{2^{3/2}}$$

$$\frac{dV}{dt} = \frac{dV}{dl} \times \frac{dl}{dt}$$

$$10 = \frac{\pi l^2}{2^{3/2}} \times \frac{dl}{dt}$$

$$\frac{dl}{dt} = \frac{10 \times 2^{3/2}}{\pi l^2}$$

$$r = 7 \quad l = 7\sqrt{2}$$

$$\therefore \frac{dl}{dt} = \frac{10 \times 2^{3/2}}{\pi (7\sqrt{2})^2} = \frac{10\sqrt{2}}{49\pi} \text{ cm/min}$$

$$= 0.092 \text{ cm/min}$$

Q. 5.

(a)  $y = 2 \cos^{-1}(4x-1)$

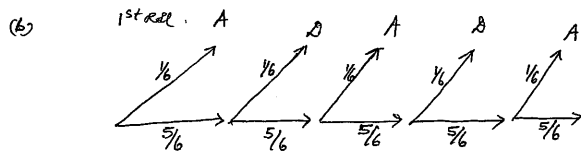
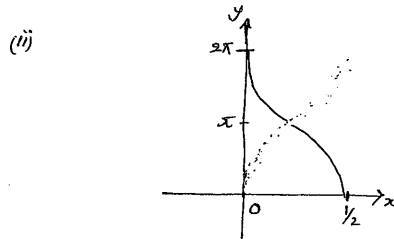
$\frac{y}{2} = \cos^{-1}(4x-1)$

(i) Domain:  $-1 \leq 4x-1 \leq 1$

$0 \leq 4x \leq 2$

$0 \leq x \leq \frac{1}{2}$

Range:  $0 \leq \frac{y}{2} \leq \pi$   
 $0 \leq y \leq 2\pi$



(i)  $P(\text{Aaron winning in the 2nd roll}) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216}$

(ii)  $P(\text{Aaron winning}) = \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \dots$

$a = \frac{1}{6} \quad r = \left(\frac{5}{6}\right)^2$

$$S = \frac{a}{1-r}$$

$$= \frac{\frac{1}{6}}{1 - \left(\frac{5}{6}\right)^2} = \frac{\frac{1}{6}}{\frac{36-25}{36}} = \frac{\frac{1}{6}}{\frac{11}{36}} = \frac{1}{6} \times \frac{36}{11} = \frac{6}{11}$$

Q. 5(c) Method 1

$2x^3 - 7x^2 + 7x - 2 = 0$

Let  $\alpha, \beta, \gamma$  be the roots.

$\therefore \frac{\beta}{\alpha} = \frac{\gamma}{\beta} = r \quad [\text{G.P.}]$

$\beta^2 = \alpha\gamma \quad \text{--- (1)}$

Also  $\alpha + \beta + \gamma = \frac{7}{2} \quad \text{--- (2)}$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{7}{2} \quad \text{--- (3)}$

$\alpha\beta\gamma = 1 \quad \text{--- (4)}$

$\alpha\beta + \beta^2 + \beta\gamma = \frac{7}{2} \quad [\text{sub (1) in (3)}]$

$\beta(\alpha + \beta + \gamma) = \frac{7}{2}$

$\beta\left(\frac{7}{2}\right) = \frac{7}{2} \quad [\text{subbing (2)}]$

$\beta = 1 \quad \checkmark$

$\therefore \alpha\gamma = 1 \quad [\text{sub } \beta=1 \text{ in (4)}]$

$\gamma = \frac{1}{\alpha}$

use this info. in (2)

$\alpha + 1 + \frac{1}{\alpha} = \frac{7}{2}$

$2\alpha^2 - 5\alpha + 2 = 0$

$(2\alpha-1)(\alpha-2) = 0$

$\alpha = \frac{1}{2} \text{ or } \alpha = 2$

When  $\alpha = \frac{1}{2} \quad \gamma = 2$

$\alpha = 2 \quad \gamma = \frac{1}{2}$

Answer:  $\frac{1}{2}, 1, 2$

or  $2, 1, \frac{1}{2}$

Method 2.

Let the roots be

$\frac{\alpha}{r}, \alpha, r\alpha$

∴  $\frac{\alpha}{r} + \alpha + r\alpha = \frac{7}{2} \quad \text{--- (1)}$

$\frac{\alpha}{r} \alpha + \frac{\alpha}{r} r\alpha + \alpha r\alpha = \frac{7}{2} \quad \text{--- (2)}$

$\frac{\alpha}{r} \times \alpha \times r\alpha = 1 \quad \text{--- (3)}$

$\alpha^3 = 1$

$\alpha = 1$

sub  $\alpha=1$  in (1)

$\frac{1}{r} + 1 + r = \frac{7}{2}$

$2r^2 - 5r + 2 = 0$

$(2r-1)(r-2) = 0$

$r = \frac{1}{2} \text{ or } r = 2$

If  $r = \frac{1}{2}$  the roots are  $2, 1, \frac{1}{2}$

If  $r = 2$  the roots are  $\frac{1}{2}, 1, 2$

5(d).

$\cos(2\sin x) = \sin(2\cos x)$

$\cos(2\sin x) = \cos(\pi/2 - 2\cos x)$

$2\sin x = 2n\pi \pm (\frac{\pi}{2} - 2\cos x) \quad \text{general solution}$

$2\sin x = 2n\pi \pm \frac{\pi}{2} \mp 2\cos x$

$2\sin x \pm 2\cos x = 2n\pi \pm \frac{\pi}{2}$

$\sin x \pm \cos x = n\pi \pm \frac{\pi}{4}$

$\sqrt{2} \sin(x \pm \frac{\pi}{4}) = n\pi \pm \frac{\pi}{4}$

$\sin(x \pm \frac{\pi}{4}) = \frac{1}{\sqrt{2}}(n\pi \pm \frac{\pi}{4})$

Q 6

$$(a) (i) \quad x^2 = 4ay \quad P(2at, at^2)$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

at  $x = 2at$  gradient  $= \frac{2at}{2a} = t$  for the tangent at P.

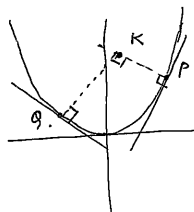
gradient of Normal is  $-\frac{1}{t}$

$\therefore$  equation of Normal (PK) is

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$y - at^2 = -\frac{x}{t} + 2a$$

$$y = at^2 + 2a - \frac{x}{t} \quad \text{--- (1)}$$



(ii) Let the point Q be  $(2aq, aq^2)$   
 $\therefore$  The gradient of Normal at Q is  $-\frac{1}{q}$ .

This normal is perpendicular to PK

$$\therefore -\frac{1}{q} \times -\frac{1}{t} = -1$$

$$\Rightarrow q = -\frac{1}{t}$$

$\therefore$  The coordinates of Q are  $(2a(-\frac{1}{t}), a(-\frac{1}{t})^2)$   
 $(-\frac{2a}{t}, \frac{a}{t^2})$ .

(iii) Find equation QK and solve QK & PK simultaneously.

$$\text{Gradient of PK} = -\frac{1}{t}$$

$$\text{Gradient of QK} = t \quad (\because \text{it } \perp \text{ to PK}).$$

Equation of QK is

$$y - \frac{a}{t^2} = t(x + \frac{2a}{t})$$

$$y = tx + 2a + \frac{a}{t^2} \quad \text{--- (2)}$$

Continued on next page:-

Q. 6 (a) cont.:-

(iii) Solving (1) & (2)

$$at^2 + 2a - \frac{x}{t} = tx + 2a + \frac{a}{t^2}$$

$$at^4 + 2at^2 - xt = xt^3 + 2at^2 + \frac{a}{t}$$

$$xt^3 + xt = at^4 - a$$

$$x(t^3 + t) = a(t^4 - 1)$$

$$x = a \frac{(t^2+1)(t^2-1)(t^2+1)}{t(t^2+1)}$$

$$= a \frac{(t^2-1)}{t}$$

$$x = a(t - \frac{1}{t})$$

Substituting this in (2)

$$y = t \cdot a(t - \frac{1}{t}) + 2a + \frac{a}{t^2}$$

$$= at^2 - a + 2a + \frac{a}{t^2}$$

$$y = at^2 + a + \frac{a}{t^2}$$

Q.6.

$$(b) (i) T = 20^\circ + 75e^{-kt}$$

$$\begin{aligned} \frac{dT}{dt} &= 75e^{-kt} \times -k \\ &= -k 75e^{-kt} \\ &= -k(T - 20) \end{aligned}$$

$$(ii) T = 20^\circ + 75e^{-kt}$$

$$75^\circ = 20^\circ + 75e^{-10k}$$

$$\frac{55}{75} = e^{-10k}$$

$$\ln \frac{11}{15} = -10k$$

$$\begin{aligned} k &= -\frac{1}{10} \ln\left(\frac{11}{15}\right) \\ &= 0.03 \end{aligned}$$

$$(iii) 60^\circ = 20^\circ + 75e^{-kt}$$

$$\frac{40}{75} = e^{-kt}$$

$$\ln\left(\frac{40}{75}\right) = -kt$$

$$\begin{aligned} t &= -\frac{1}{k} \ln\left(\frac{40}{75}\right) \\ &= \frac{\ln(40/75)}{\frac{1}{10} \ln(11/15)} \\ &= 20.26 \text{ minutes} \end{aligned}$$

Lawrie should arrive on or before

10:40:17 am.

Q.7

(a)

$$(i) x = vt \cos \theta$$

$$y = -\frac{1}{2}gt^2 + vt \sin \theta$$

$y=0$  when the projectile returns to ground at  $Z$

$$\Rightarrow vt \sin \theta = \frac{1}{2}gt^2$$

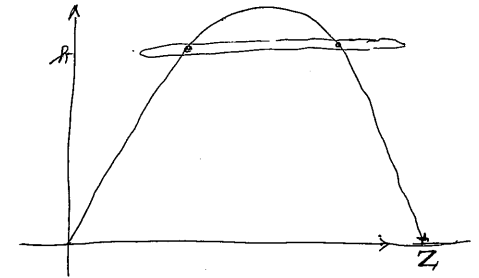
$$t(v \sin \theta - \frac{1}{2}gt) = 0$$

$$\therefore t=0$$

$$\text{OR } v \sin \theta - \frac{1}{2}gt = 0$$

$$\therefore t = \frac{2v \sin \theta}{g}$$

$$\begin{aligned} \text{And } Z = x &= v \left( \frac{2v \sin \theta}{g} \right) \cos \theta \\ &= \frac{v^2 2 \sin \theta \cos \theta}{g} \\ &= \frac{v^2 \sin 2\theta}{g} \end{aligned}$$



$$(ii) \text{ when } y = h$$

$$h = -\frac{1}{2}gt^2 + vt \sin \theta$$

$$2h = -gt^2 + 2vt \sin \theta$$

$$gt^2 - 2vt \sin \theta + 2h = 0 \quad [\text{Quadratic in } t]$$

$$t = \frac{2v \sin \theta \pm \sqrt{4v^2 \sin^2 \theta - 4g2h}}{2g} \quad \text{--- (1)}$$

to pass through the clouds twice  $t$  must have two distinct values.

$$\therefore 4v^2 \sin^2 \theta - 4g2h > 0$$

$$v^2 \sin^2 \theta - 2gh > 0$$

$$v^2 \sin^2 \theta > 2gh$$

$$h < \frac{v^2 \sin^2 \theta}{2g}$$

Q.7 cont. ....  
(a)

(iii) From equation (1) in part (ii)

$$t = \frac{2v \sin \theta \pm \sqrt{4v^2 \sin^2 \theta - 4g \cdot h}}{2g}$$

$$t = \frac{v \sin \theta \pm \sqrt{v^2 \sin^2 \theta - 2gh}}{g}$$

$$t = \frac{320 \times \sqrt{3}/2 \pm \sqrt{320 \times 3/4 - 2 \times 10 \times 3000}}{10}$$

$$= \frac{277.13 \pm 129.6}{10}$$

$$t_1 = 14.8 \quad t_2 = 40.67$$

(iv)  $x_1 = 320 \times 14.8 \times \cos 60$

$$x_2 = 320 \times 40.67 \times \cos 60$$

$$x_2 - x_1 = 320 \times \cos 60 (40.67 - 14.8)$$

$$= 4,139.2 \text{ m.}$$

Q.7 (b) (i)  $\ddot{x} = \frac{dv}{dx} \left( \frac{1}{2} v^2 \right) = 32x^3 - 48x^2 + \frac{16x^2}{x}$

Integrating  $\frac{1}{2} v^2 = 8x^4 - 16x^3 + 8x^2 + C$

$$v^2 = 16x^4 - 32x^3 + 16x^2 + C$$

$$24^2 = 16 \times 3^4 - 32 \times 3^3 + 16 \times 3^2 + C$$

$$x=3, v=24$$

$$C=0$$

$$\therefore v^2 = 16x^4 - 32x^3 + 16x^2$$

$$v^2 = 16x^2 (x^2 - 2x + 1)$$

$$v^2 = 16x^2 (x-1)^2 \quad \checkmark$$

(ii)  $\therefore v = \pm 4x(x-1)$  [since  $x=3, v=24$ ]  
 $v = 4x(x-1)$

$$\frac{dx}{dt} = v \Rightarrow \frac{dt}{dx} = \frac{1}{v} \quad \int \frac{dt}{dx} dx = \int \frac{1}{4x(x-1)} dx \quad \rightarrow$$

$$t = \int \frac{1}{4x(x-1)} dx$$

$$\therefore \int \frac{1}{x(x-1)} dx = 4t \quad \checkmark$$